CSA 0919 PROGRAMMING IN JAVA

EASY LEVEL QUESTION 31-55

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Main.java

```java
import java.util.*;
class Main{
 static int gcd(int a,int b){return b==0?a:gcd(b,a%b);}
 static int lcm(int a,int b){return a/gcd(a,b)*b;}
 public static void main(String[] args){
  Scanner sc=new Scanner(System.in);
  int n=sc.nextInt(),g=0,l=1;
  for(int i=0;i<n;i++){
   int x=sc.nextInt();
   g=(i==0)?x:gcd(g,x);
   l=lcm(l,x);
  }
  System.out.println("LCM="+l);
  System.out.println("GCD="+g);
 }
}
```

input

```
20
LCM=80
GCD=4
```

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Main.java

```java
import java.util.*;
class Main{
 static double si(double p,int y,boolean s){
  return (p<=0||y<=0)?0:p*y*(s?12:10)/100;
 }
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  double p=sc.nextDouble();
  int y=sc.nextInt();
  boolean s=sc.next().equalsIgnoreCase("y");
  System.out.println("Interest="+si(p,y,s));
 }
}
```

input

```
3
n
Interest=60000.0
```

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Main.java

```java
import java.util.*;
class Main{
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int n=sc.nextInt(),x=0,y=1;
  for(int i=0;i<n;i++){
   System.out.print(x+" ");
   int t=x+y; x=y; y=t;
  }
 }
}
```

input

```
5
0 1 1 2 3 5
```

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```java
import java.util.*;
class Main{
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int n=sc.nextInt(),x=0,y=1,sum=0;
  for(int i=0;i<=2*n;i++){
   if(i%2==0) sum+=x;
   int t=x+y; x=y; y=t;
  }
  System.out.println(sum);
 }
}
```

input

```
4
33
```

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Main.java

```java
import java.util.*;
class Main{
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int m=sc.nextInt(),n=sc.nextInt(),k=sc.nextInt();
  for(int i=m;i<=n;i+=k+1)
   System.out.print(i+" ");
 }
}
```

input

```
50
100
7
50 58 66 74 82 90 98
```

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Main.java

```java
import java.util.*;
class Main{
 static boolean comp(int n){
  if(n<=1) return false;
  for(int i=2;i<n;i++)
   if(n%i==0) return true;
  return false;
 }
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int x=sc.nextInt(),y=sc.nextInt();
  for(int i=x;i<=y;i++)
   if(comp(i)) System.out.print(i+" ");
 }
}
```

input

```
12
19
12 14 15 16 18
```

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Main.java

```java
import java.util.*;
class Main{
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int n=sc.nextInt();
  long f=1;
  for(int i=1;i<=n;i++) f*=i;
  System.out.println("Factorial="+f);
 }
}
```

```
4
Factorial=24
```

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Main.java

```java
import java.util.*;
class Main{
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int y=Integer.parseInt(sc.next().split("/")[2]);
  System.out.println((y%4==0&&y%100!=0)||y%400==0?
   "Leap Year":"Non Leap Year");
 }
}
```

input

```
04/11/1947
```

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Main.java

```java
import java.util.*;
class Main{
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int n=sc.nextInt(),c=0;
  for(int i=1;i<=n;i++)
   if(n%i==0)c++;
  System.out.println("Factors="+c);
 }
}
```

```
100
Factors=9
```

40

Main.java

```java
import java.util.*;
class Main{
 public static void main(String[] a){
  Scanner sc=new Scanner(System.in);
  int n=sc.nextInt(),s=0;
  for(int i=1;i<n;i++)
   if(n%i==0)s+=i;
  System.out.println(s==n?
   "Perfect Number":"Not Perfect");
 }
}
```

```
6
Perfect Number
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double n = sc.nextDouble();

        System.out.println("Square Number: " + (n * n));
        System.out.println("Cube Number: " + (n * n * n));
    }
}
```

input

```
0.6
Square Number: 0.36
Cube Number: 0.216
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++)
            System.out.print((2*i - 1) + " ");
    }
}
```

input

```
3
1 3 5
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int temp = n, sum = 0;

        while(temp > 0){
            int d = temp % 10;
            sum += d*d*d;
            temp /= 10;
        }

        if(sum == n)
            System.out.println("Given number is Armstrong number");
        else
            System.out.println("Not Armstrong number");
    }
}
```

input

```
153
Not Armstrong number
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String num = sc.next();
        int sum = 0;

        for(char c : num.toCharArray())
            sum += c - '0';

        while(sum > 9){
            int temp = 0;
            while(sum > 0){
                temp += sum % 10;
                sum /= 10;
            }
            sum = temp;
        }

        System.out.println("Final Sum: " + sum);
    }
}
```

input

```
143
Final Sum: 8
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int r = (int)Math.sqrt(n);

        if(r*r == n)
            System.out.println("Square Root: " + r + ", " + (-r));
        else
            System.out.println("Not a Perfect Square");
    }
}
```

input

```
6561
Square Root: 81, -81
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        if(n <= 0){
            System.out.println("false");
            return;
        }

        while(n % 3 == 0)
            n /= 3;

        System.out.println(n == 1);
    }
}
```

input

```
27
true
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        String paragraph = "Ram hit a ball the hit ball flew far after it
        String[] banned = {"hit"};

        Set<String> ban = new HashSet<>(Arrays.asList(banned));
        Map<String,Integer> map = new HashMap<>();

        for(String word : paragraph.toLowerCase().split(" ")){
            if(!ban.contains(word))
                map.put(word, map.getOrDefault(word,0)+1);
        }

        String result = Collections.max(map.entrySet(),
                Map.Entry.comparingByValue()).getKey();

        System.out.println(result);
    }
}
```

input

```
ball
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        int[] arr = {1,0,2,3,0,4,5,0};

        for(int i=0;i<arr.length;i++){
            if(arr[i]==0 && i<arr.length-1){
                for(int j=arr.length-1;j>i;j--)
                    arr[j]=arr[j-1];
                i++;
            }
        }

        System.out.println(Arrays.toString(arr));
    }
}
```

input

```
[1, 0, 0, 2, 3, 0, 0, 4]
```

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```java
class Main {
    public static void main(String[] args) {
        int[] nums = {3,0,1};
        int n = nums.length;
        int sum = n*(n+1)/2;

        for(int x : nums)
            sum -= x;

        System.out.println(sum);
    }
}
```

```
2
```

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```java
class Main {
    public static void main(String[] args) {
        int[] nums = {-2,1,-3,4,-1,2,1,-5,4};
        int max = nums[0], cur = 0;

        for(int x : nums){
            cur = Math.max(x, cur + x);
            max = Math.max(max, cur);
        }

        System.out.println("Maximum Sum: " + max);
    }
}
```

```
Maximum Sum: 6
```

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```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int m = sc.nextInt();
        int n = sc.nextInt();

        for(int i=1;i<=n;i++)
            System.out.println(i+"x"+m+"="+(i*m));
    }
}
```

```
3x4=12
4x4=16
5x4=20
```

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Main.java

```java
class MyThread extends Thread {
    public void run(){
        System.out.println("Thread Running");
    }
}
class Main {
    public static void main(String[] args) throws Exception {
        MyThread t = new MyThread();
        Thread.sleep(500);
        t.start();
        t.join();
        System.out.println("Main Thread End");
    }
}
```

input

```
Thread Running
Main Thread End
```

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Main.java

```java
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int ch;
        do{
            System.out.println("1.If 2.For 3.While 0.Exit");
            ch = sc.nextInt();

            if(ch==1) System.out.println("Syntax: if(condition){}");
            if(ch==2) System.out.println("Syntax: for(init;cond;inc){}");
            if(ch==3) System.out.println("Syntax: while(condition){}");

        }while(ch!=0);
    }
}
```

input

```
1.If 2.For 3.While 0.Exit
```

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Main.java

```java
import java.util.*;
class Gen<T extends Comparable<T>> {
    void sort(T[] arr){
        Arrays.sort(arr);
    }
}
class Main {
    public static void main(String[] args) {
        Integer[] a = {3,1,2};
        new Gen<Integer>().sort(a);
        System.out.println(Arrays.toString(a));
    }
}
```

input

```
[1, 2, 3]
```

55

```java
public class Main {

    void add(int a, int b) {
        System.out.println("Sum = " + (a + b));
    }

    void add(int a, int b, int c) {
        System.out.println("Sum = " + (a + b + c));
    }

    public static void main(String[] args) {
        Main obj = new Main();
        obj.add(5, 10);
        obj.add(5, 10, 15);
    }
}
```